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The liberalization of marijuana laws may have implications for neighborhood crime insofar as the distribution of marijuana through a dispensary system may provide additional opportunities for criminal behavior to take place. This project fills an important gap in the scant literature on medical marijuana dispensaries and neighborhood crime rates by integrating perspectives from environmental criminology and social organization theories in investigating the dispensary-crime nexus through interaction models and flexibly assessing dispensaries' relationship to crime at different spatial scales. This study found the placement of a medical marijuana dispensary in the previous year to be associated with crime rate change, in both the block and the surrounding area, over and above predictor variables drawn from social organization theory. And, this study's interaction models suggest that marijuana dispensaries may increase crime rates on socially organized blocks, with such blocks potentially experiencing a slight perturbation in their ecological continuity from a dispensary's establishment.

Keywords neighborhoods and crime; drug policy; environmental criminology; social organization; theoretical integration

In 1996, the voters of the State of California passed Proposition 215, which essentially liberalized California's marijuana laws by permitting the possession, cultivation, and distribution of marijuana for medicinal purposes. The passage of Proposition 215 made California the first US state to medicalize

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marijuana. In turn, given the nascent industry that accompanied the drug's medicalization, ambiguities arose regarding the meaning of the law's provisions. As a result, in 2003, California legislators enacted Senate Bill 420, which, aside from clarifying the provisions of Proposition 215, established a dispensary system. This system grants the formation of *dispensaries*, which operate for the purpose of distributing the drug to authorized patients, who have received a recommendation from a physician. In other words, dispensaries are facilities, which are nested within neighborhoods and distribute marijuana. However, the establishment of medical marijuana dispensaries in neighborhoods is not without controversy. In 2009, the California Police Chiefs Association's Task Force on Marijuana Dispensaries released a white paper on dispensaries, arguing anecdotally that these facilities produce adverse secondary effects, with attendant increases in the homicide and robbery rates, in the neighborhoods where they are located (California Police Chiefs Association, 2009).

Since the publication of this white paper on medical marijuana dispensaries, researchers have conducted studies on dispensaries to assess the veracity of the claims set forth by this white paper. In California cities, such as Los Angeles, which is the study site for the current paper, medical marijuana dispensaries tend to be situated in poor, minority neighborhoods that are also surrounded by alcohol outlets (Morrison, Gruenewald, Freisthler, Ponicki, & Remer, 2014; Thomas & Freisthler, 2015). Given that dispensaries necessarily make marijuana more readily available, unsurprisingly, marijuana use, abuse, and dependency tend to accompany the placement of dispensaries in Californian communities (Freisthler & Gruenewald, 2014; Mair, Freisthler, Ponicki, & Gaidus, 2015). Current marijuana use and marijuana dispensary density are also found to be related to child physical abuse (Freisthler, Gruenewald, & Wolf, 2015). Therefore, social problems, of which crime rate increases are just one, may accompany the establishment of medical marijuana dispensaries in neighborhoods. The liberalization of marijuana laws may have implications for neighborhood crime insofar as the distribution of marijuana through a dispensary system may provide additional opportunities for criminal behavior to take place. Few empirical studies have examined the link between medical marijuana dispensaries and neighborhood crime rates: with one examining the relationship cross-sectionally at the tract-level (Kepple & Freisthler, 2012), another also cross-sectionally at the facility-level (Freisthler, Kepple, Sims, & Martin, 2013), and still further one examining the relationship longitudinally across a 24-month period at the block group level (Freisthler, Ponicki, Gaidus, & Gruenewald, 2016). The present study is one of few longitudinal studies to assess the spatial relationship between medical marijuana dispensaries and neighborhood crime rates.

I add to the scant literature on the relationship between medical marijuana dispensaries and neighborhood crime in the following ways. First, I take my analyses to an appropriately sized spatial unit, the census block (Kim, 2016), for examining these arguably micro-spatial processes.[1] Indeed, the unit of aggregation used for neighborhood studies of crime ought to be driven by theory and the research question at hand (Hipp, 2007). Researchers conducting state-level analyses of the relationship between states' enactment of medical marijuana laws and violent and property crime rates have found either no relationship or a negative one with the enactment of such laws on crime (Morris, TenEyck, Barnes, & Kovandzic, 2014; Shepard & Blackley, 2016); however, the findings from such studies cannot generalize to the processes that may take place with the establishment of dispensaries in neighborhoods. Second, I assess the spatial extent of the dispensary-crime nexus by employing a spatial buffer measure, which captures blocks within a half-mile radius of the focal block. Such a modeling strategy, of including a spatially lagged version of the main independent variable, enables one to test whether the processes that may link these facilities to crime operate on a larger geographic scale (Taylor, 2015). Third, following up on recent work on the shifting ecologies of crime (Taylor, Ratcliffe, & Perenzin, 2015), I estimate crime rate change as a function of a dispensary's placement in the previous year. In capturing crime rate change through a lagged regression framework, this study also provides a more stringent test of the relationship between neighborhood facilities and crime compared to studies viewing a cross-sectional relationship. Lastly, I examine whether indicators of social organization, such as average household income and percent homeowners, condition the association between medical marijuana dispensaries and neighborhood rates of crime. On the one hand, higher socioeconomic and more residentially stable blocks with dispensaries may fend off possible crime rate increases; the placement of a dispensary on these blocks may activate residents' guardianship potential (Sampson, Raudenbush, & Earls, 1997). On the other hand, the placement of marijuana dispensaries may provide neighborhood opportunities for criminal behavior to take place on socially organized blocks. These facilities may produce disorderly conditions, which in turn reduce resident-based control of socially organized blocks, subjecting such blocks to possible crime rate increases (California Police Chiefs Association, 2009). In looking solely at general effects, researchers may be ignoring certain contextual factors that may moderate these facilities' relationship to crime.

Theoretical Perspectives and Related Literature

Environmental Criminology

Following earlier scholarship examining criminogenic facilities and crime, I situate my discussion within an environmental criminology theoretical

1. This statement should not be taken as implying that "smaller is better" in communities and crime research.

framework. I argue that medical marijuana dispensaries in the City of Los Angeles may be crime attractors insofar as they draw in motivated offenders into the neighborhoods where these dispensaries are found; dispensaries generally lack access to banking (Chemerinsky, Forman, Hopper, & Kamin, 2015), making ripe for criminal victimization these facilities' patrons, who carry cash on their person. Given that federal law prohibits financial institutions from allowing "marijuana-related businesses to open accounts, the marijuana industry largely operates on a cash-only basis—a situation that attracts thieves" (Hill, 2015, p. 597). Consequently, greater convergences between motivated offenders and suitable targets may take place (Cohen & Felson, 1979). Based on the logic of the environmental criminology school of thought, I maintain that medical marijuana dispensaries, like other criminogenic facilities, pattern crime opportunities along the paths that neighborhood residents, patrons from the wider community, and motivated offenders take (Brantingham & Brantingham, 1993). As an influx of people are drawn into the neighborhoods where these dispensaries are found, the social milieu of these neighborhoods subsequently enters into the awareness space of offenders, who then come to familiarize themselves with the neighborhood setting and therefore integrate such a setting into their routine activities. Because medical marijuana dispensaries are generally a cash-only economy that distributes a drug that is one of the most widely used substances and among the top cash crops in the United States (Caulkins, Kilmer, & Kleiman, 2016), neighborhoods with these facilities may become hot spots of predatory crime (Sherman, Gartin, & Buerger, 1989).

Other researchers looking at potentially criminogenic facilities and neighborhood crime also ground their thinking in the environmental criminology tradition (Haberman & Ratcliffe, 2015). Environmental criminologists maintain that crime, much like ordinary activities that characterize one's day (e.g. shopping and driving home from work), follows certain rhythms and tempos, which are in turn shaped by the distribution of these criminogenic facility types across the city landscape, providing criminal opportunities for offenders as they go about their routine activities (Kinney, Brantingham, Wuschke, Kirk, & Brantingham, 2008). Accordingly, a criminal event unfolds if and only if motivated offenders and suitable targets converge in both time and space in the absence of capable guardians (Cohen & Felson, 1979). Neighborhood facilities provide an easy conduit through which these offender-victim convergences take hold insofar as the placement of these facilities in neighborhoods draw in from the wider community outsiders, who may be criminally motivated or may have no psychic attachment to the neighborhoods hosting these various amenities. The placement of these facilities in neighborhoods may, therefore, lead to higher crime rates (Brantingham & Brantingham, 1993).

These facilities may be classified as either crime generators or crime attractors. On the one hand, crime generators are particular places that draw in large numbers of people, many of whom may be motivated offenders and still others suitable targets. Though offenders who frequent these places may not necessarily be criminally motivated at the outset, when presented with opportunities they will exploit such opportunities, which are in abundance given the increased foot traffic accompanying these sorts of places. To make more concrete this concept, these places are usually commercial establishments, such as malls, restaurants, and grocery stores (Brantingham & Brantingham, 2008). On the other hand, crime attractors are "particular places, areas, neighborhoods, districts which create well-known criminal opportunities to which strongly motivated, intending criminal offenders are attracted because of the known opportunities for particular types of crime" (Brantingham & Brantingham, 1995, p. 8). As such, they usually are attractive because they are places that are cash economies. These places include bars, payday lenders, pawnshops, drug markets, and prostitution areas.

Researchers examining the association between criminogenic facilities and neighborhood crime have largely assessed the average effects of these facilities (Bernasco & Block, 2011; Kurtz, Koons, & Taylor, 1998; Wo, 2016). Looking at calls for service data from a Philadelphia inner-city neighborhood, Kurtz and colleagues (1998) found that storefronts, which draw in large crowds of people from outside the focal neighborhood, undermined residential control of the area, increased physical disorder in the surrounding area, and resulted in more calls for police service. Estimating negative binomial regression models with longitudinal data, Wo (2016) found that alcohol outlets and banking institutions increase both violent and property crime rates in census tracts containing these facilities. A rich body of literature, mostly grounded in the field of public health, finds that alcohol outlet density is a robust predictor of aggravated assault rates (Nielsen & Martinez, 2003; Pridemore & Grubesic, 2012, 2013). Another body of literature finds that fringe banks (i.e. payday lenders, pawnshops, and check-cashing centers), despite being disproportionately situated in poor, minority neighborhoods increase crime rates, over and above neighborhood structural context measures drawn from social organization theory (Kubrin & Hipp, 2016; Kubrin, Squires, Graves, & Ousey, 2011). Whereas previous research on criminogenic facilities and crime tends to focus merely on the general effects of these various facilities, the current study adds to this literature by positing that medical marijuana dispensaries' relationship to crime may be conditioned by community context.

Social Organization Theory

I also frame my discussion through a social organization theoretical framework, arguing for a move away from just examining the average effects of potentially criminogenic facilities, and instead also taking into account the social context. Community context may matter in understanding the ecological association between medical marijuana dispensaries and neighborhood crime rates. Like fringe banks (Kubrin & Hipp, 2016), medical marijuana dispensaries in Los Angeles may be disproportionately situated in precarious, minority-occupied neighborhoods wherein low socioeconomic status and residential instability characterize these communities. The neighborhood context of where medical marijuana dispensaries are placed may have implications for crime rates to the extent that residents who live in less socially organized neighborhoods may lack the capable guardianship to effectively combat crime (Sampson et al., 1997). Whereas environmental criminology typically focuses more on the convergence between offenders and victims, social organization theory takes a more explicit focus on the presence or absence of capable guardianship (Reynald, 2010; Weisburd et al., 2016). I argue that capable guardianship may either be activated or deactivated on socially organized blocks with a newly established dispensary.[2] So, by exploring interactions between neighborhood structural characteristics and medical marijuana dispensaries, one can better contextualize how interactions may play out between criminal opportunity and micro-place guardianship.

Theoretically, then, two perspectives may account for why community contextual factors condition dispensaries' relationship to neighborhood crime rates. One perspective holds that social organization—broadly defined as the ability of neighborhood residents to realize their common values and resolve local problems (Kubrin & Weitzer, 2003)—attenuates dispensaries' association with crime. According to this perspective, socially organized neighborhoods have more guardianship potential to prevent crime around dispensaries. So, the placement of a dispensary on a socially organized block may serve as a cue for residents to activate capable guardianship, which in turn thwarts offender-victim convergences from taking place on such blocks. In other words, socially organized blocks may be resilient to the external shock that accompanies the placement of a marijuana dispensary because they may be better equipped to fend off possible crime rate increases (Stokols, Lejano, & Hipp, 2013).

Another perspective holds that medical marijuana dispensaries established on socially organized blocks may engender disorderly conditions, which in turn reduce residents' sense of territorial control, leading to an increase in crime. As the California Police Chiefs Association (2009) put it simply in their white paper, "Marijuana dispensaries bring in the criminal element and loiterers" (p. 14). Socially organized blocks with a recently established dispensary may experience an influx of outsiders, who "may litter, write graffiti, or simply cause more wear and tear on a block" (Kurtz et al., 1998, p. 123). Residents in turn may perceive more neighborhood incivilities, fear crime, and feel less in control over their community (Innes, 2004). This shift in the social dynamics of such blocks may lead to a deactivation of capable guardianship on the part of residents, who may perceive a risk to their personal safety (Reynald, 2010). As a result, socially organized communities may instead provide neighborhood opportunities for criminal behavior to take place. So, the placement of

2. I thank two of the reviewers for raising points regarding moderating mechanisms. I also thank a reviewer for highlighting the theoretical import of guardianship at the nexus of dispensaries and crime.

medical marijuana dispensaries on socially organized blocks may be associated with crime rate increases.

Some researchers have begun to explore possible interaction effects between criminogenic facilities and neighborhood structural resources, such as socioeconomic status and residential stability, on crime rates (Kubrin & Hipp, 2016; Peterson, Krivo, & Harris, 2000; Pridemore & Grubesic, 2012). These variables tapping into neighborhood structural resources serve as proxies for social organization (see Sampson, Morenoff, & Gannon-Rowley, 2002). In creating such interaction terms between criminogenic facilities and neighborhood structural characteristics, some researchers have attempted to theoretically integrate environmental criminology and social organization perspectives (Rice & Smith, 2002; Smith, Frazee, & Davison, 2000). On the one hand, Pridemore and Grubesic (2012), using a social organization index measure, found that social organization weakens alcohol outlets' association with crime in Cincinnati, Ohio neighborhoods. On the other hand, Kubrin and Hipp (2016) found a positive association between residential stability and crime rates for Los Angeles city blocks with a fringe bank. They concluded that "the usual protective effect of homeowners on crime rates is not evident for blocks with a fringe lender" (p. 778). The present study adds to the literature on medical marijuana dispensaries and crime by integrating environmental criminology and social organization perspectives through the creation of interaction terms between dispensaries and neighborhood structural characteristics, such as socioeconomic status and residential stability.

Although some research has tested whether the effects of potentially criminogenic facilities on crime are moderated by neighborhood structural characteristics, no research has tested this for medical marijuana dispensaries. In fact, only a few studies have investigated possible linkages between dispensaries and neighborhood crime in general, regardless of the context. Using environmental criminology as a framework for their study, Kepple and Freisthler (2012) employed a cross-sectional, ecological research design to assess whether a possible link exists between dispensaries and property and violent crime in census tracts. Although they found no such association, they only controlled for crime opportunity proxy measures, thereby neglecting community contextual factors in their statistical models. And, most importantly, like Kepple and Freisthler (2012), Freisthler et al. (2013) also did not take into account the community context in which these dispensaries are situated. Freisthler and colleagues (2013) did a premise survey of medical marijuana dispensaries in Sacramento, California, coding for security measures; they then drew a series of buffers around these dispensaries to examine the extent to which crime clustered around dispensaries. They found that such security measures "might be effective at reducing crime within the immediate vicinity of the dispensaries" (p. 278), although plausibly they do not after a certain distance away from these establishments. Ratcliffe (2012) found that, with alcohol outlets, crime did not necessarily cluster in the immediate vicinity but did after a certain distance. Yet, as discussed above, the community context of neighborhoods hosting these facilities may be consequential for either mitigating or exacerbating the relationship between dispensaries and crime. Through exploring possible interactions between neighborhood structural characteristics and these facilities, perspectives in social organization theory and environmental criminology can be integrated.

The Spatial Extent of Facility Influence

Only recently have researchers begun to examine the spatial extent of potentially criminogenic facilities. Ratcliffe (2012), for example, estimated change-point regression models to assess bars' relationship to crime and found that a threshold effect takes hold whereby violent crime clusters spatially within 85 feet of bars but then diminishes significantly thereafter. In examining the spatial extent of facilities' influence on crime, interestingly, Groff and Lockwood (2014) found that the spatial relationship between facilities and crime varies depending on facility type. On the one hand, researchers studying criminogenic facilities and crime largely find the association between these facilities and crime to be micro-spatial (Bernasco & Block, 2011; Groff, 2014; Groff & Lockwood, 2014; Ratcliffe, 2012). On the other hand, medical marijuana dispensaries may have a larger spatial relationship to crime. Given that social problems such as use, dependency, and child physical abuse are associated with the location of these facilities (Freisthler & Gruenewald, 2014; Freisthler et al., 2015; Mair et al., 2015), such facilities' relationship to crime may have a broader spatial extent than other types of facilities.

In sum, medical marijuana dispensaries may be crime attractors because they are generally cash economies that distribute the most widely used illicit substance. As a result, dispensaries may draw in motivated offenders onto the blocks where these facilities are located. Marijuana dispensaries in Los Angeles may attract a criminal element to the communities where they are established, with offenders preying on patrons of these establishments and residents. Dispensaries' relationship to interpersonal crime may be more pronounced on the block, given the propinquity of social contact between offenders and victims, who may carry cash or marijuana on their person. Their relationship to acquisitive crime may be broader in spatial scale, for offenders' awareness space may extend to nearby blocks as a result of a dispensary's placement in the wider community. And, dispensaries' association with crime may also be conditioned by the community context of Los Angeles city blocks.

Data and Methodology

Data were compiled on medical marijuana dispensaries, crime, land use, alcohol outlets, and socio-demographic characteristics of Los Angeles city communities to examine the dispensary-crime nexus. Los Angeles, California, was used as the study area for these analyses. Los Angeles may be a suitable study area for examining criminogenic facilities and crime. It is representative of a large US urban city, with a population that greatly exceeds one million people, of which about half are non-white. Moreover, it contains a large number of medical marijuana dispensaries, which at one point surpassed the number of Starbucks in Los Angeles city neighborhoods (Hecht, 2014). Given the constant establishment of these facilities, the City of Los Angeles may be an appropriate research area to test whether the placement of a dispensary on a block is associated with crime rate increases in both the focal block and the area surrounding the block. The general research strategy is to test the relationship between the placements of marijuana dispensaries in 2013 on the change in crime from 2013 to 2014.

The census block is the unit of analysis. The average residential block in Los Angeles is .017 square miles, which is about the size of eight American football fields (Goodell, 2016). Given that communities and crime researchers emphasize the theoretical import of choosing an appropriately sized spatial unit for one's analyses (Boessen & Hipp, 2015; Hipp, 2007; Kim, 2016), the present study uses the census block as the unit of analysis. Criminogenic facilities and crime researchers typically use much smaller spatial units, such as census blocks (Bernasco & Block, 2011; Haberman & Ratcliffe, 2015; Kubrin & Hipp, 2016), for the processes that may link these facilities to crime are theorized as operating on a smaller geographic scale. This choice of using a smaller spatial unit, such as the census block, is informed by environmental criminology, which refers to crime generators and crime attractors as micro-places (see Bernasco & Block, 2011, p. 36). It follows that the census block is an appropriately sized spatial unit to employ in using environmental criminology as a framework for assessing whether medical marijuana dispensaries in Los Angeles are crime attractors.

Variables

The dependent variables for this study are 2014 counts of violent (i.e. homicide, robbery, and aggravated assault) and property crimes (i.e. burglary, larceny, and motor vehicle theft) coded and reported by the Los Angeles Police Department for the Federal Bureau of Investigation's annual Uniform Crime Reports. The official crime data were cleaned, geocoded, and aggregated to the census block.

The main independent variable is a medical marijuana dispensary count measure. This measure was created using a listing of businesses categorized as Medical Marijuana Collectives, which registered with the Los Angeles Office of Finance, obtained a business tax registration certificate (BTRC), and paid their annual taxes. The Los Angeles Office of Finance is a city agency responsible for the collection of revenue from businesses operating in the city. This listing of Medical Marijuana Collectives was obtained from the Office of Finance on February 2015. These data include the names of dispensaries, where they are located, and the date on when they were established. I geocoded the addresses of medical marijuana dispensaries and then aggregated the data to the census block. I restricted my analyses only to dispensaries established in 2013 to examine the relationship between dispensaries' placement during 2013 and crime rate change from 2013 to 2014. I also included a spatial buffer measure of the count of the number of dispensaries within a half-mile of the block, with an inverse distance decay function, to account for the surrounding area. I tested buffers of various sizes and found that a half-mile spatial buffer measure fit better with the data.[3] To create this spatially lagged version of the main independent variable, I first created a spatial weights matrix. This entailed defining as neighbors, using a distance decay function, all blocks within a half-mile radius of the focal block, with such blocks expected to affect the focal block, but with inverse distance decay.[4] I then multiplied this matrix by the values in nearby blocks for dispensaries to construct this spatial buffer measure (see Boggess & Hipp, 2016).[5]

The social organization variables (i.e. socioeconomic status and residential stability) were downloaded directly from the 2010 US Census FTP website for blocks (http://www2.census.gov/census_2010/04-Summary_File_1/), except for average household income, which is only available at the block group level. The average household income measure, which operationally defines the *socioeconomic status* construct in social organization theory, accounts for the socioeconomic context of neighborhoods. Average household income values for blocks were imputed through a prediction model that uses block group data to predict these values. This regression-based imputation model necessarily constrains the block values for average household income to add up to the total at the block group level, accounts for the uncertainty in this prediction through its standard errors, and multiply imputes such values (see Boessen & Hipp, 2015).[6] The *residential stability* construct in social organization theory is captured with the percentage of homeowners. These measures will then be used, separately, to create a marijuana dispensary-social organization interaction term by multiplying the medical marijuana dispensary count measure with each of these measures.[7] These interaction terms incorporate perspectives from both environmental criminology and social organization theory.

3. I assessed the spatial extent of the dispensary-crime nexus using the following buffer sizes: 400 feet, 800 feet, 1/4 mile, 1/3 mile, 2/5 mile, and 1/2 mile. I assessed goodness of fit with the pseudo R-squared statistic.

4. An inverse distance decay function implies that nearer blocks are more influential than more distant blocks (Rengert, Piquero, & Jones, 1999).

5. This same procedure was done for the creation of other spatially lagged independent variables. These spatial buffer measures guard against potential error in the geo-references in the police data that are greater in size than a census block, for they account for the area surrounding the block.

6. Please refer to this online supplemental material which describes in depth this procedure: http://onlinelibrary.wiley.com/wol1/doi/10.1111/1745-9125.12074/suppinfo

7. Average household income and percent homeowners tap into social organization theory's related concept of *structural resources*, which may be consequential for engendering capable guardianship on blocks (Sampson, 2008). Yet, such resources may provide additional opportunities for offenders frequenting blocks with a dispensary (Cohen & Felson, 1979).

Control variables include the following measures. I control for the proportion of blocks with industrial, office, residential, and retail land use, with other land use as the reference category, using 2012 land use data from the Southern California Association of Governments; bars and liquor stores; percent vacant units; percent black; percent Latino; and ethnic heterogeneity. The demographic control measures were downloaded directly from the 2010 US Census for blocks. Ethnic heterogeneity is operationally defined through the Herfindahl Index, which is based on five racial/ethnic groupings (i.e. white, black, Latino, Asian, and other races) on blocks (Gibbs & Martin, 1962). The measures for bars and liquor stores were created using 2012 North American Industry Classification System codes for bars (722410) and liquor stores (445310) from ReferenceUSA, a national source for business and consumer data. To account for the surrounding area, I also included half-mile spatial buffer measures for average household income, percent homeowners, percent black, percent Latino, ethnic heterogeneity, percent vacant units, bars, and liquor stores.

Analytic Strategy

Given that the outcome measures are crime counts and are over-dispersed, I estimated negative binomial regression models. I included block population as the exposure variable, effectively making the outcomes interpretable as rates (Osgood, 2000).[8] In the two sets of models, I included the measures enumerated above, but I also controlled for logged crime rates of the previous year. It follows that these models captured the percentage change in the crime rate from 2013 to 2014. The idea is to explain crime rate change explicitly, with such change potentially related to the placement of a medical marijuana dispensary in the previous time point. Theoretically, then, crime rate change can be referred to as *ecological discontinuity*, with it being "captured in a lagged regression framework" (Taylor, 2015, p. 153) and possibly associated with the crime-opportunity structure accompanying a dispensary's establishment on a block. The first set of models estimated the relationship between the six crime types in 2014 and medical marijuana dispensaries established in 2013 on the block and the surrounding half-mile area, controlling for socio-demographic characteristics of the block and the surrounding half-mile area, land use characteristics of the block, alcohol outlets on the block and in surrounding area, and crime rates of the previous year. The second set of models included interaction terms between medical marijuana dispensaries on the block and the neighborhood structural characteristics of average household income and

8. Blocks with zero population were necessarily dropped from the analyses, given the inclusion of socio-demographics measures in the negative binomial regression models (see Bernasco & Block, 2011).

percent homeowners. Interaction terms between medical marijuana dispensaries in the area surrounding the block and block measures for neighborhood structural characteristics were also included in this final set of models.

A demographic portrait of blocks with and without marijuana dispensaries warrants discussion. As one can see from Table 1, blocks with marijuana dispensaries tend to be lower income, Latino, less residentially stable, and surrounded by alcohol outlets. This is consistent with previous research examining the location of dispensaries in Californian communities (Morrison et al., 2014; Thomas & Freisthler, 2015). Such blocks may be characterized as less socially organized, whereas those without dispensaries may be classified as more organized. Although blocks without a dispensary may not be characteristic of high-crime communities, they are, on average, surrounded by two dispensaries within a half-mile radius. The spatial relationship between dispensaries and crime may extend to such blocks.

Table 1 Descriptive statistics of variables used in analyses, split by blocks with and without marijuana dispensaries

	No dispensaries		Has dispensaries	
	Mean	*SD*	Mean	*SD*
Blocks				
Average household income	90,077	73,153	54,798	44,657
Percent homeowners	55.04	33.24	30.31	28.10
Ethnic heterogeneity	.41	.20	.42	.21
Percent black	10.30	19.60	12.01	20.24
Percent Latino	39.99	32.14	48.15	31.69
Percent vacant units	5.76	7.34	7.44	9.68
Marijuana dispensaries	0	0	1.08	.29
Proportion industrial land use	.03	.11	.02	.11
Proportion office land use	.02	.07	.02	.05
Proportion residential land use	.75	.33	.76	.31
Proportion retail land use	.06	.13	.06	.13
Bars	.01	.12	.10	.38
Liquor stores	.02	.15	.14	.39
Surrounding 1/2 mile				
Average household income	81,808	53,648	61,015	23,802
Percent homeowners	45.84	24.39	32.08	19.28
Ethnic heterogeneity	.48	.16	.50	.15
Percent black	9.52	14.93	10.39	14.95
Percent Latino	42.79	28.14	48.44	26.10
Percent vacant units	6.29	3.05	7.11	2.92
Marijuana dispensaries	2.32	4.27	4.40	5.94
Bars	4	10.11	8.75	19.14
Liquor stores	6.77	7.33	10.06	8.44

Results

Before discussing the results from the models, I present my results from assessing the level of spatial autocorrelation. I assessed spatial autocorrelation in the residuals of the models after including the spatial buffers and found no evidence of spatial clustering of the residuals. Whereas the Moran's I values for the crime types were .01 for homicide, .08 for robbery, .10 for aggravated assault, .06 for burglary, .02 for larceny, and .07 for motor vehicle theft, the Moran's I values for the residuals of the models were all .01 or less. It follows that the models accounted for much of the spatial clustering in the crime data. In other words, the residuals assessed how much clustering remained after taking into account this explicit spatial modeling approach.

I begin my discussion with the set of models that includes the count of the number of medical marijuana dispensaries on the block and the spatial buffer measure for dispensaries in the half-mile area surrounding the block. As shown in Table 2, and consistent with environmental criminology, the placement of medical marijuana dispensaries on a block is positively related to five of the six crime types. Hence, these models show that a 245% increase in the homicide rate from 2013 to 2014 is associated with an additional dispensary in 2013 on a block $[\exp(\beta) - 1]100$, after accounting for alcohol outlets and the sociodemographic characteristics of both the block and the surrounding area. What is more, these models show 49, 44, 38, and 22% increases in the rates for robbery, aggravated assault, larceny, and motor vehicle theft, respectively, associated with each additional dispensary on a block. These models also suggest that the spatial relationship between dispensaries and crime may extend to the half-mile area surrounding the block. Three and four percent increases in the rates for aggravated assault, and burglary and larceny, respectively, are related to a one standard deviation increase in dispensaries established in the surrounding area in 2013.

Social Organization at the Intersection of Dispensaries and Crime

In the final set of analyses, I examined whether social organization conditions the association between medical marijuana dispensaries and neighborhood crime rates. I assessed this by including interaction terms between medical marijuana dispensaries in the block and the surrounding area, respectively, and block measures for average household income and percent homeowners.[9] I present significant interactions in these models, which are displayed in Table 3. Average household income and percent homeowners, respectively, appear to condition the association between medical marijuana dispensaries and crime rates in both the block and the surrounding area. The results from interaction

9. In the Figures to be presented below, all plotted values are present in the data.

Table 2 Predicting crime rate change in blocks using measures in blocks and the surrounding area

	Homicide	Robbery	Aggravated assault	Burglary	Larceny	Motor vehicle theft
Block						
Marijuana dispensaries	1.2380**	.4011**	.3657**	.0842	.3190**	.1961**
	(3.8894)	(3.3889)	(3.0999)	(1.1195)	(4.6925)	(2.7723)
Average household income	−.0094	−.0114*	−.0138**	−.0014	−.0044*	−.0020
	(−.4353)	(−2.3518)	(−3.1022)	(−.6583)	(−2.0032)	(−.8764)
Percent homeowners	−.0056	−.0102**	−.0071**	−.0018**	−.0042**	−.0032**
	(−1.3353)	(−11.7145)	(−8.5860)	(−4.2488)	(−9.9379)	(−7.2921)
Ethnic heterogeneity	.6731	.5355**	.2861*	.2114**	.3036**	.2121**
	(.9607)	(3.8049)	(2.0680)	(3.0030)	(4.4126)	(2.8468)
Percent black	.0117	−.0017	.0059**	.0027*	.0028*	.0016
	(1.1859)	(−.8349)	(2.9570)	(2.2080)	(2.4530)	(1.3160)
Percent Latino	.0052	.0014	.0049**	−.0032**	.0007	−.0003
	(.6351)	(.9565)	(3.4326)	(−4.4370)	(1.0522)	(−.3453)
Percent vacant units	.0219*	.0075**	.0130**	.0011	.0037**	.0034*
	(2.1313)	(3.1979)	(5.7609)	(.7651)	(2.9116)	(2.4096)
Proportion industrial use	−.2373	.1100	−.0152	.0028	.1241	.0297
	(−.2710)	(.5855)	(−.0868)	(.0320)	(1.4322)	(.3297)
Proportion office use	.4426	.3633	.3036	.3226*	.0968	.1799
	(.4240)	(1.4541)	(1.2723)	(2.4954)	(.7268)	(1.3521)
Proportion residential use	−.0423	−.0641	−.0487	.0501	.0675*	.0561
	(−.1502)	(−.9517)	(−.7884)	(1.4494)	(1.9635)	(1.6063)
Proportion retail use	.0496	.1248	−.0758	.1442†	.2396**	.0315
	(.0757)	(.8399)	(−.5305)	(1.9014)	(3.1920)	(.4040)
Bars	.5500	.3539**	.4954**	.1535**	.2290**	.2135**
	(1.5366)	(3.9161)	(5.4667)	(2.8318)	(4.5325)	(4.0312)
Liquor stores	.0765	.3501**	.2828**	.0959*	.2189**	.0750†
	(.2273)	(4.5464)	(3.8578)	(2.0801)	(5.0659)	(1.6773)

Surrounding area (.5 mile)						
Marijuana dispensaries	.0044	.0034	.0116**	.0046*	.0035†	.0001
	(.2998)	(.9341)	(3.3528)	(2.2296)	(1.6669)	(.0524)
Average household income	.0424	−.0746**	−.0508**	−.0017	−.0060	−.0204**
	(.5941)	(−5.1411)	(−4.1080)	(−.4079)	(−1.4717)	(−4.1451)
Percent homeowners	−.0155*	−.0061**	−.0008	−.0045**	−.0008	−.0028**
	(−2.0857)	(−3.9727)	(−.5664)	(−6.4913)	(−1.1484)	(−3.9259)
Ethnic heterogeneity	.5348	−.4444*	−.1173	−.0547	−.3226**	−.0224
	(.5421)	(−2.2228)	(−.6186)	(−.5767)	(−3.4463)	(−.2259)
Percent black	.0352**	.0184**	.0122**	.0027†	−.0023†	.0020
	(2.9002)	(7.2463)	(5.0809)	(1.9127)	(−1.6932)	(1.3437)
Percent Latino	.0329**	.0058**	.0080**	−.0007	−.0018*	.0039**
	(2.8381)	(2.8248)	(4.0860)	(−.7580)	(−2.0144)	(4.0881)
Percent vacant units	−.0004	.0167*	.0302**	−.0048	.0097**	−.0075†
	(−.0082)	(2.1130)	(4.2476)	(−1.1952)	(2.6148)	(−1.7283)
Bars	.0035	.0046**	.0021	.0024*	.0017†	.0002
	(.4562)	(2.9990)	(1.4246)	(2.5592)	(1.9208)	(.2582)
Liquor stores	−.0016	.0053*	.0059**	.0021	.0004	−.0029*
	(−.1701)	(2.1935)	(2.6099)	(1.5218)	(.2730)	(−2.0618)
Previous crime rate (logged)						
2013 Homicide rate	.5135**					
	(5.5219)					
2013 Robbery rate		.9602**				
		(58.1592)				

(*Continued*)

Table 2 (*Continued*)

	Homicide	Robbery	Aggravated assault	Burglary	Larceny	Motor vehicle theft
2013 Aggravated assault rate			.6982**			
			(4.4075)			
2013 Burglary rate				.7306**		
				(77.8153)		
2013 Larceny rate					.9093**	
					(112.9702)	
2013 Motor vehicle theft rate						.7223**
						(73.2347)
Intercept	−10.0465**	−1.6109**	−3.4666**	−1.5842**	−.9569**	−1.8533**
	(−8.1098)	(−6.7591)	(−15.6905)	(−16.7735)	(−10.4712)	(−17.6982)
N	22,655	22,655	22,655	22,655	22,655	22,655

Notes. $^{**}p < .01$; $^{*}p < .05$; $^{\dagger}p < .10$. *T*-values in parentheses. Negative binomial regression models.

Table 3 Interactions

	Robbery	Aggravated assault	Burglary	Larceny	Motor vehicle theft
Dispensaries × income	.0657*	.0407	.0318†	−.0055	.0321†
	(2.1570)	(1.2826)	(1.7770)	(−.3215)	(1.8523)
Dispensaries × owners	.0098*	.0107*	.0022	.0036	.0050†
	(2.1321)	(2.3652)	(.8024)	(1.3937)	(1.8583)
Dispensaries (1/2 mile) × income	.0014	.0011	.0013**	.0000	.0013**
	(1.6326)	(1.3654)	(3.2162)	(.0369)	(2.9421)
Dispensaries (1/2 mile) × owners	.0001	.0000	.0000	.0001	.0002**
	(.4518)	(.2724)	(.5343)	(.9643)	(2.8441)
N	22,655	22,655	22,655	22,655	22,655

Notes. $^{**}p < .01$; $^{*}p < .05$; $^{\dagger}p < .10$. *T*-values in parentheses. Same controls as main effect models in Table 2 but not shown.

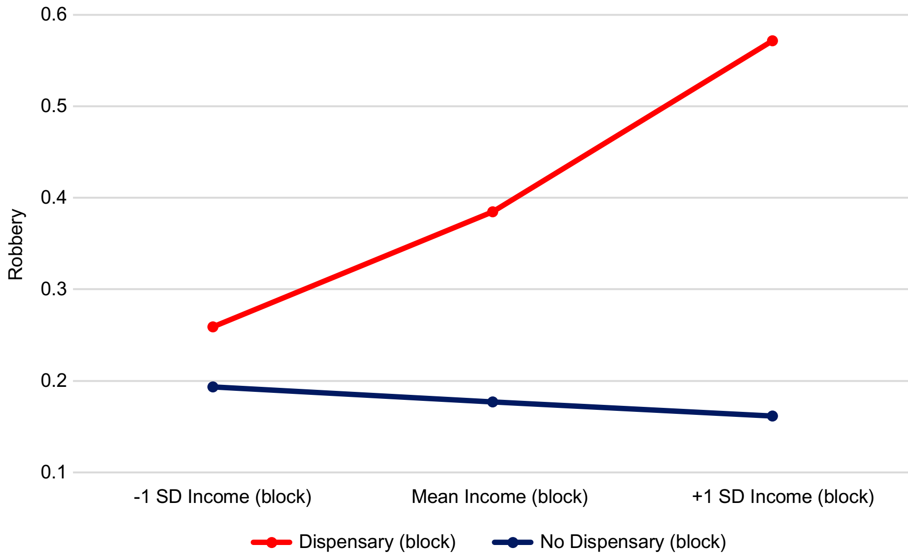

Figure 1 Robbery: interaction of marijuana dispensaries and average household income.

models using income as a moderator warrants mention. These results imply that socio-economic status may serve as an important moderator in the relationship between medical marijuana dispensaries and neighborhood crime rates insofar as it could provide additional opportunities for offenders.

To highlight further these findings, I graphed these interactions between dispensaries and average household income. I first look at how income moderates the relationship between dispensaries in the block and robbery rates. Figure 1 is a graph plotting the predicted robbery rate from the model for a block with, and a block without, a marijuana dispensary from one standard deviation below the mean to one standard deviation above the mean for average household income, which is the moderating variable. This figure shows that, on the one hand, there is essentially no relationship between income in the block and the robbery rate when there is no dispensary (the bottom line), given that the line is essentially flat. On the other hand, this relationship is large and positive for blocks with a marijuana dispensary (the top line). It appears that income does not protect blocks with a marijuana dispensary against crime. Instead, higher income blocks may provide additional criminal opportunities for offenders.

On the other hand, Figure 2 plots the predicted burglary rate for blocks with one standard deviation below the mean to one standard deviation above the mean in marijuana dispensaries *in the surrounding area* and average household income from one standard deviation below the mean to one standard deviation above the mean in the focal block. First, this figure indicates that the number of marijuana dispensaries in the area surrounding the block may not matter for the burglary rate for low income blocks, given that all three lines are basically

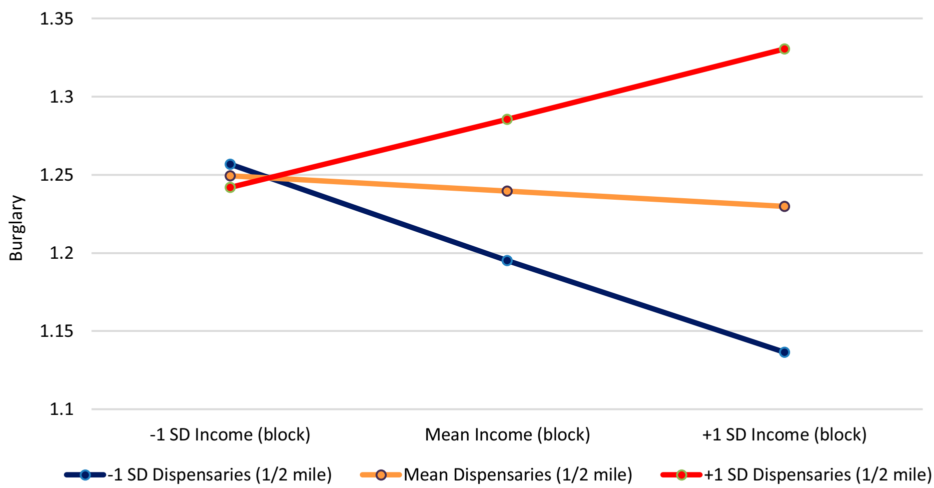

Figure 2 Burglary: interaction of marijuana dispensaries within 1/2 mile and average household income within block.

lying on top of each other on the left side of the figure. This figure shows that, on the one hand, there is a negative relationship between income and the burglary rate for blocks when there are no dispensaries in the surrounding area. On the other hand, this figure shows that this relationship is positive for blocks with a high number of dispensaries in the surrounding area. So, high income blocks with a greater number of dispensaries in the surrounding area are expected to have a higher burglary rate than blocks surrounded by few dispensaries. In other words, income appears not to be protective against crime for blocks with a high number of dispensaries in the surrounding area. Rather, such blocks may fall along the paths that offenders take moving to and from dispensaries, which provide additional criminal opportunities for offenders. Although not shown, the interaction between average household income in the block and dispensaries in the surrounding area showed a similar pattern for the motor vehicle theft rate as in Figure 2 for the burglary rate. Indeed, motor vehicle theft rates increase as both income in the block and dispensaries in the surrounding area increase.

Although not presented, homeownership on the block shows a slightly different interaction with dispensaries in the surrounding area to the extent that high homeownership blocks generally have fewer motor vehicle thefts. But, those with a high number of dispensaries in the surrounding area have more such crimes than similarly high homeownership blocks with essentially no dispensaries in the surrounding area.

Similarly, homeownership moderates the block-level relationship between dispensaries and robbery and aggravated assault rates. Although the interaction plots are not displayed, they show that blocks with a medical marijuana dispensary have higher rates of these serious crimes, homeownership notwithstanding, given that the lines do not cross on these plots. For robbery, while there is a negative relationship between percent homeowners and the robbery rate for blocks without a dispensary, there is no such relationship for blocks with a dispensary. Rather, blocks with a medical marijuana dispensary have higher robbery rates, percent homeowners notwithstanding, given that the slope is essentially flat for such blocks. For aggravated assault, blocks with a dispensary have higher aggravated assault rates, homeownership notwithstanding. Moreover, for blocks without a dispensary, there is a negative relationship between percent homeowners and assault; but, for blocks with a dispensary, this relationship is positive. These results imply that residentially stable blocks with a dispensary may experience increases in violence following the establishment of a dispensary.

Discussion and Conclusion

Neighborhoods with potentially criminogenic facilities tend to have higher crime rates, net of known correlates of crime (Bernasco & Block, 2011; Haberman & Ratcliffe, 2015; Pridemore & Grubesic, 2013). These facilities relate to crime either through a generation of foot traffic or through the attraction of a certain criminal element to their premises. The liberalization of marijuana laws may link to neighborhood crime rates insofar as the distribution of marijuana through a dispensary system may provide additional opportunities for criminal behavior to take place. In the context of the City of Los Angeles, the association between medical marijuana dispensaries and crime rates may be tied to dispensaries' lack of banking access and regulation, necessarily making them a cash economy and possibly inviting a criminal element to blocks hosting these facilities.[10] As such, these facilities in Los Angeles may be characterized as crime attractors, for they may draw in motivated offenders, who in turn prey on patrons of these establishments and residents in the wider community.

The results from this study suggest that medical marijuana dispensaries in the City of Los Angeles may be crime attractors. The placement of marijuana dispensaries in 2013 is related to the percentage change in crime rates from 2013 to 2014, over and above predictor variables drawn from social organization theory. In particular, the establishment of a medical marijuana dispensary on a block in 2013 links to change in violent crime rates from 2013 to 2014, controlling for neighborhood structural characteristics, land use, demographics, and alcohol outlets. This relationship is most pronounced for the homicide rate; this is consistent with the expectation that robberies committed on a block with a dispensary might go awry, leading to a homicide incident.

10. Some gap exists between the law on the books and the law in action (Calavita, 2016). In spite of federal law generally making banking access difficult, some US states, such as Arizona, have tried to forge an un-prosecutable relationship between banks and dispensary owners (Stern, 2014). Findings from the present study are limited to the context of the City of Los Angeles.

Unsurprisingly, then, this relationship is also strong for the robbery rate; this lends support to the proposition that patrons of these facilities might be at a higher risk for this violent victimization, for they carry cash on their person. Still further, increases in aggravated assault rates link to blocks with a newly established dispensary. Increases in larceny and motor vehicle theft rates are also associated with the establishment of dispensaries on a block. This study adds to previous research on medical marijuana dispensaries and crime by taking its analyses to the block, which may be an appropriately sized spatial unit for examining these arguably micro-spatial processes, and finds a robust ecological association between dispensaries and neighborhood crime rates.

The results from this study also suggest that dispensaries' spatial relationship to crime may be broader in scope inasmuch as increases in the rates for aggravated assault, burglary, and larceny were coupled with the establishment of a dispensary in the surrounding area. Whereas previous researchers studying criminogenic facilities and crime have largely theorized these facilities' relationship to crime to be micro-spatial (Bernasco & Block, 2011; Groff, 2014; Groff & Lockwood, 2014; Ratcliffe, 2012), medical marijuana dispensaries in Los Angeles are such that their spatial relationship to crime may operate on a larger geographic scale. This was anticipated given that research in the field of social work has found the placement of dispensaries in Californian communities to be related to various social problems, such as use, abuse, and dependency (Freisthler & Gruenewald, 2014; Freisthler et al., 2015; Mair et al., 2015). As such, these findings are consistent with Groff and Lockwood's (2014) recent work on the variable spatial relationship between criminogenic facilities and crime to the extent that "[t]he geographic extent of a facility's criminogenic influence varies by type of facility" (p. 278), with the dispensary-crime nexus possibly encompassing a larger, macro-level area (e.g. a half-mile area or greater).

Like earlier work by Smith and his colleagues attempting to integrate environmental criminology and social organization perspectives (Rice & Smith, 2002; Smith et al., 2000), this study's findings also suggest that integrating perspectives from both environmental criminology and social organization theories through interaction models may be a worthwhile endeavor. However, unlike this earlier work, in the current study, indicators of social organization (i.e. structural resources such as income and homeownership) tap into the notion of guardianship, which may interact with criminal opportunity; this implies theoretical integration. This study's interaction models suggest that medical marijuana dispensaries may relate to crime rate increases on socially organized blocks. Socially organized blocks may experience a slight perturbation in their ecological continuity from a dispensary's establishment, either on the block or in the surrounding area. Indicators of social organization, such as high socioeconomic status and residential stability, do not necessarily attenuate the association between medical marijuana dispensaries and neighborhood crime rates. Instead, blocks with more structural resources (i.e. higher income and homeownership) are connected to crime rate increases following the establishment of a dispensary in the previous year. Whereas researchers previously have found socially organized neighborhoods with criminogenic facilities such as alcohol outlets to be linked with lower violent crime rates (Pridemore & Grubesic, 2012), in the case of dispensaries, this appears not to be the case for socially organized blocks in the current study. These findings are consistent with Kubrin and Hipp's (2016) recent work on fringe banks and crime on Los Angeles city blocks wherein they found that more residentially stable blocks with a fringe bank were associated with higher crime rates. Although fringe banks, like medical marijuana dispensaries, in Los Angeles are also disproportionately located on less socially organized blocks, these facilities may be tied to crime rate increases on more socially organized blocks, as well. And, more socially organized blocks without a dispensary may still link to property crime rate increases, if surrounded by dispensaries. This study moves the medical marijuana dispensaries and neighborhood crime literature towards the theoretical importance of social context moderating the dispensary-crime nexus at different spatial scales.

This study is not without limitations. First, because this study's area consisted of only one major US city, results are not necessarily generalizable to other US cities. Therefore, future research should examine whether the dispensary-crime nexus holds across other US cities. Second, given that there is no existing licensing agency for dispensaries, this study's listing of marijuana businesses may not capture all dispensaries opening up in 2013 in Los Angeles; it only captures those that registered as businesses and paid their annual taxes. Still, this data source has mapped onto estimates of the number of dispensaries operating in Los Angeles, with BTRC renewals reported by the Office of Finance matching very closely with the number of dispensaries reported by researchers (Romero, 2015; UCLA Medical Marijuana Research, 2015). Third, given that official crime data were used for this study, underreporting on the part of crime victims necessarily accompanied the use of such data. Nevertheless, criminologists find under-reporting on the part of crime victims to be unrelated to neighborhood characteristics (Baumer, 2002). Fourth, this study did not account for dispensaries' security features, such as security guards and cameras, which may be consequential for crime prevention. Still, such features have been found to matter only within the immediate premises of dispensaries (Freisthler et al., 2013). Fifth, given that this study only examined dispensaries established in the same year and their relationship to crime rates the following year, findings from this study cannot speak to the temporal nature of the dispensary-crime nexus. Future research would need to examine longitudinally the association that the organizational age of marijuana dispensaries has with neighborhood crime rates (see Wo, Hipp, & Boessen, 2016).

Marijuana law and policy warrants discussion. At the time of this writing, four US states (i.e. Alaska, Colorado, Oregon, and Washington) have legalized the commercial production and distribution of marijuana for adult use (National Organization for the Reform of Marijuana Laws, 2016). Anticipating marijuana's likely legalization via ballot initiative in 2016 (see DiCamillo, 2016), the State of California established a statewide licensing and regulatory system for medical marijuana businesses, with this system expected to be operable by 2018 and to serve as a template for a legalized marijuana industry. However, unless cities, such as Los Angeles, create their own permit system for these businesses as expected by state law, marijuana dispensaries will continue to be unregulated drug market places; the Los Angeles Times' editorial board went so far as to say that "The City Council should not let marijuana businesses set city policy" (The Times Editorial Board, 2016; para. 6). If such businesses continue to operate within a "gray area" of legality, they may come to be seen as locally undesirable land uses (LULUs), creating stigma and reinforcing perceptions of disorder for places with these facilities (Sampson & Raudenbush, 2004). Unlike LULUs, however, marijuana businesses provide cities millions of dollars in revenue, making them otherwise conventional businesses (Németh & Ross, 2014). And, these business regulation issues are important, given that marijuana's likely legalization in California may be a tipping point for legalization across the United States (Caulkins et al., 2016).

Yet, in spite of state marijuana laws being increasingly liberalized, public safety concerns may still arise due to marijuana dispensaries' potential cash-based element. This cash-based element of marijuana dispensaries may result from dispensaries' general lack of access to banking (Hill, 2015). As some legal scholars put it, "the core of the banking problem is the continuing illegality of marijuana at the federal level" (Chemerinsky et al., 2015, p. 93). One resolution of this issue may entail the federal government not prosecuting banks and credit unions providing financial services to marijuana businesses complying with state law; another may involve the US Congress outright removing marijuana as a scheduled drug per the Controlled Substances Act (1970). This would require US cannabis policy moving beyond stalemate (Room, Fischer, Hall, Lenton, & Reuter, 2010). In the absence of such sweeping change at the federal level, other policy options are available for state and local governments with these legalized drug markets. Home delivery of marijuana to users may serve as a viable option to the storefront dispensary. City governments could also require dispensary owners to better target-harden their premises or employ third-party policing (see Buerger & Mazerolle, 1998). Whether dispensaries establish on more or less socially organized blocks, directed police patrol on blocks with a dispensary is also in order. Such targeted law enforcement efforts would reduce crime, not just on blocks with a newly established dispensary, but also in the surrounding area. A diffusion of crime control benefits would follow necessarily from such a course of action (Taniguchi, Rengert, & McCord, 2009; Weisburd & Green, 1995; Weisburd et al., 2016).

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